Universal Stability of Structural Observables Across Dimensions:

A Comparative Test of CCI, Structural Free Energy, and Entropy–Information Scaling

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Abstract

We investigate the dimensional robustness of several structural observables in nonlinear ϕ^4 field systems across one, two, and three spatial dimensions. The tested quantities are the Critical Coherence Index (CCI), structural free energy F_{struct}, mutual information (MI), positive entropy production $\dot{S}^{(+)}$, and the entropy–information ratio used in earlier scaling analyses.

The goal is to determine which quantities remain comparatively stable under dimensional change, and which behave as projection-dependent approximations. We find that the entropy–information ratio undergoes a dramatic dimensional blow-up, increasing by nearly three orders of magnitude from 1D to 3D, whereas CCI and F_{struct} remain within a comparatively controlled range. This instability is not merely quantitative, but structural: the ratio ceases to define a consistent observable across dimensions. This supports the interpretation that ratio-based scaling relations are not fundamental, but instead represent lower-dimensional projections of a deeper structural description.

The results strengthen the role of CCI and structural free energy as robust diagnostics for nonlinear field dynamics, while suggesting that entropy–information scaling should be treated as an effective phenomenological proxy rather than a universal law.

Keywords: critical coherence index, structural free energy, nonlinear field theory, entropy production, mutual information, dimensional robustness, scaling laws

1 Introduction

A central question in the study of nonlinear field systems is whether the observables used to characterise order, instability, and selection remain meaningful when the dimensionality of the system changes. In previous work, the Critical Coherence Index (CCI) was introduced as a dimensionless structural diagnostic for distinguishing ordered, critical, and chaotic regimes in nonlinear ϕ^4 field dynamics [8] [1, 4]. A related empirical result was the entropy–information scaling relation, in which the CCI could be approximated by a ratio involving positive entropy production and structural mutual information [2].

However, subsequent studies in higher dimensions revealed a tension: while the monotonic relation between entropy production, information, and CCI persists, the effective exponent description becomes degenerate in 3D [3]. This raises an important interpretive issue: is the entropy–information ratio itself a fundamental structural object, or is it merely a projection of a deeper selection geometry?

To address this question, we perform here a direct cross-dimensional stability test. We compare the distributions and mean values of five key observables across 1D, 2D, and 3D field systems:

- the Critical Coherence Index (CCI),
- the structural free energy F_{struct},

- mutual information (MI),
- positive entropy production $\dot{S}^{(+)}$,
- and the entropy–information ratio.

The aim is not to prove strict universality, but to evaluate which quantities remain comparatively stable under dimensional extension, and to identify those that behave as projection-dependent approximations.

2 Method

We use previously generated datasets from 1D, 2D, and 3D ϕ^4 simulations, computed using the same definitions and normalisations as in the earlier papers of this series [1, 2, 3]. For each dimension, we extract the following mean observables:

$$\text{CCI}, \qquad F_{\text{struct}}, \qquad I_{\text{nn}}, \qquad \dot{S}^{(+)}, \qquad R_\alpha. \tag{1}$$

Here:

- CCI is the Critical Coherence Index,
- F_{struct} is the structural free energy,
- I_{nn} is nearest-neighbour mutual information [7],
- $\dot{S}^{(+)}$ is positive coarse-grained entropy production [6],
- R_α denotes the best-fit entropy–information ratio extracted from the scaling analysis.

To assess dimensional robustness, we compare:

1. the mean value of each observable in 1D, 2D, and 3D,
2. the variance of these mean values across dimensions,
3. the relative drift defined by
$$D_{\text{rel}} = \frac{\max(\mu_d)}{\min(\mu_d)}, \tag{2}$$
where μ_d is the mean value in dimension d.

A small value of D_{rel} indicates that the observable remains comparatively stable across dimensions, while large values indicate strong dimensional sensitivity.

3 Results

The measured cross-dimensional means are shown in Table 1. The associated distribution plots are shown in Figure 1.

Table 1: Cross-dimensional means and relative drift of the tested observables.

Observable	**1D Mean**	**2D Mean**	**3D Mean**	**Max/Min Drift**
CCI	0.303	0.457	0.601	1.98
F_{struct}	0.430	0.641	0.824	1.92
MI	0.574	0.449	0.393	1.46
$\dot{S}^{(+)}$	0.263	0.080	0.111	3.30
Ratio	6.81	71.64	13563.11	1992

The most striking feature is the extreme dimensional instability of the entropy–information ratio. Its mean value increases from 6.81 in 1D to 1.36×10^4 in 3D, corresponding to a relative drift of approximately 1992. This behaviour sharply contrasts with the CCI and structural free energy, whose relative drifts remain below 2.

Mutual information shows only moderate change, while entropy production itself varies more strongly, especially between 1D and 2D. This suggests that the blow-up of the ratio is not caused by a single observable alone, but by the instability of the ratio construction under dimensional change.

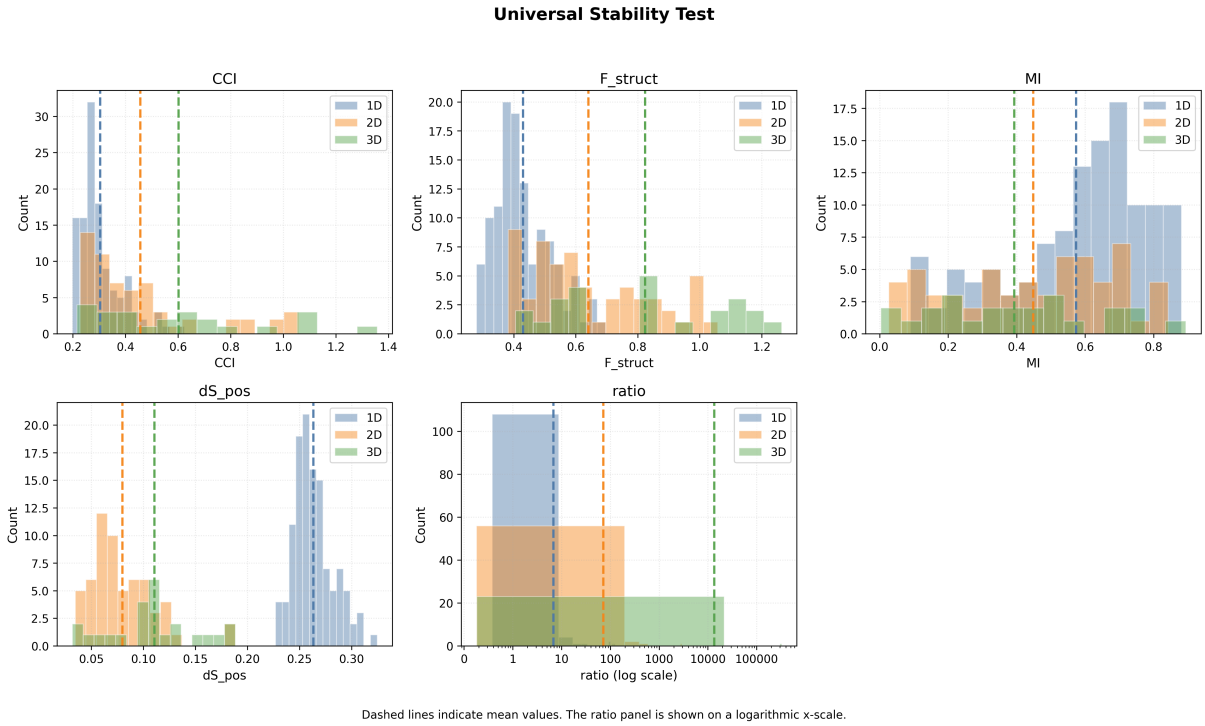

Figure 1: Universal stability test across 1D, 2D, and 3D ϕ^4 systems. Shown are the distributions of the Critical Coherence Index (CCI), structural free energy, mutual information, positive entropy production, and the entropy–information ratio. Dashed lines indicate the corresponding mean values. The ratio panel is displayed on a logarithmic x-scale ($\log_{10}$). The CCI and F_{struct} remain comparatively controlled across dimensions, whereas the ratio exhibits strong dimensional distortion, especially in 3D.

4 Interpretation

The results support a clear hierarchical interpretation of the tested quantities.

4.1 CCI as a comparatively robust structural observable

The CCI remains within a comparatively narrow range across dimensions, with a relative drift of 1.98. While this is not zero—and we do not claim strict dimensional invariance—it suggests that the CCI captures a structural property of the underlying dynamics that is not strongly tied to a specific projection or dimensional representation. This supports the interpretation that the CCI defines a comparatively intrinsic coordinate on the space of dynamically realised configurations, in the sense developed in [3].

4.2 Structural free energy as a stable global effective quantity

F_{struct} also remains relatively stable, with a drift of 1.92, slightly stronger than the CCI. This is consistent with its role as a global effective selection functional: it remains meaningful across systems, but naturally reflects increased structural complexity in higher dimensions.

4.3 Entropy–information scaling as a structurally unstable proxy

The entropy–information ratio behaves qualitatively differently. Its dimensional blow-up is not merely quantitative: the ratio ceases to define a consistent observable across dimensions. This indicates that it cannot be treated as a universal structural quantity. Instead, it functions as a lower-dimensional effective proxy whose numerical behaviour depends strongly on how entropy production and mutual information project the underlying dynamics onto a particular dimensional representation.

This interpretation is fully consistent with the broader picture developed in the earlier papers: the monotonic structure survives, but simple exponent-based or ratio-based descriptions become structurally unstable as dimensionality rises.

5 Discussion

These results refine the interpretation of entropy–information scaling. While entropy production has proven useful as a diagnostic of nonequilibrium phase transitions in general settings [5, 6], the present results show that ratio-based combinations of entropy production and mutual information [7] become structurally unstable under dimensional extension in ϕ^4 systems. The earlier empirical success of ratio-based descriptions in 1D and 2D should not be read as evidence for a fundamental law of the form

$$\text{CCI} \sim \frac{\dot{S}^{(+)}}{I^{\alpha}}. \tag{3}$$

Rather, such relations should be interpreted as effective low-dimensional approximations to a deeper structural geometry.

In contrast, the CCI itself remains comparatively robust, suggesting that it may be closer to an intrinsic coordinate on the solution manifold. This supports the interpretation that structural observables (CCI, F_{struct}) define intrinsic coordinates on the solution manifold, whereas scaling ratios correspond to dimension-dependent projections of that geometry. Likewise, structural free energy continues to behave as a meaningful global effective description of selection behaviour.

What can be claimed. The present analysis supports the following cautious conclusion:

> CCI and structural free energy are substantially more dimensionally robust than ratio-based scaling quantities. The instability of the ratio is not merely quantitative, but structural: it ceases to define a consistent observable across dimensions.

What cannot yet be claimed. The present results do *not* establish strict universality of the CCI. To make such a claim, one would need to test additional model classes beyond ϕ^4 systems, such as sine-Gordon, reaction–diffusion, or Kuramoto–Sivashinsky dynamics. The observed drift of ~ 2 for the CCI across dimensions is moderate but non-trivial, and should be treated as partial rather than complete dimensional robustness.

6 Conclusion

We have performed a direct cross-dimensional stability test of the main observables used in the structural selection programme. The results reveal a strong contrast between compara-

tively robust structural diagnostics (CCI and F_{struct}) and structurally unstable scaling proxies (entropy–information ratios).

The key finding is that the entropy–information ratio undergoes extreme dimensional drift, while the CCI and structural free energy remain within a comparatively controlled range. This supports a hierarchical interpretation in which:

- CCI acts as a comparatively robust structural observable,
- F_{struct} acts as a stable global effective functional,
- ratio-based scaling acts as a structurally unstable, projection-dependent approximation.

These findings strengthen the conceptual move from simple scaling laws toward a more general structural selection geometry, and provide direct empirical support for interpreting CCI and F_{struct} as intrinsic coordinates on the solution manifold rather than as effective scaling proxies.

Reproducibility

All numerical analyses and figures presented in this paper were produced using a single self-contained Python script. The script loads the three simulation datasets (1D, 2D, and 3D), computes global means, variance, and relative drift for each observable, and generates Figure 1 with dashed mean lines and a logarithmic x-scale for the ratio panel.

Script. The full analysis script is publicly available at:

https://github.com/Chris4081/structural-selection-principle/blob/main/uni_stability_test.py

Data. The three input datasets and the scripts used to generate them are:

CSV file	Generated by
`cci_entropy_information_test.csv`	`python cci_alpha_scaling.py`
`2d_cci_entropy_information_test.csv`	`python cci_entropy_scaling_2d.py`
`3d_cci_entropy_information_test.csv`	`python cci_entropy_scaling_3d.py`

All datasets and generation scripts are available in the same repository.

Dependencies. The script requires standard scientific Python libraries: `pandas`, `numpy`, `matplotlib`, and `scipy`. No custom packages or external data sources are needed. All observables are computed using the same definitions and normalisations as in the earlier papers of this series [1, 2, 3].

References

[1] C. Krieg, *A Structural Free Energy on the Solution Manifold: Coherence Diagnostics, Critical Coherence Index, and Regime Classification*, preprint (2026).

[2] C. Krieg, *Entropy–Information Scaling of the Critical Coherence Index in Nonlinear Field Dynamics*, preprint (2026).

[3] C. Krieg, *Scaling Laws as Projections: Evidence for a Dimension-Dependent Scaling Manifold in Nonlinear Field Systems*, preprint (2026).

[4] C. Krieg, *Predictive Power of the Critical Coherence Index: From Structural Diagnostics to Regime Classification*, preprint (2026).

[5] C. E. F. Noa, P. E. Harunari, M. J. de Oliveira, and C. E. Fiore, *Entropy production as a tool for characterizing nonequilibrium phase transitions*, Phys. Rev. E **100**, 012104 (2019).

[6] T. Tomé and M. J. de Oliveira, *Entropy production in nonequilibrium systems at stationary states*, Phys. Rev. Lett. **108**, 020601 (2012).

[7] P. Kowalek and B. Schroer, *Relative Entropy and Mutual Information in Gaussian Statistical Field Theory*, Ann. Henri Poincaré (2024), DOI: 10.1007/s00023-024-01522-2.

[8] H. Kleinert and V. Schulte-Frohlinde, *Critical Properties of ϕ^4-Theories*, World Scientific, Singapore (2001).